

Projection Operators in Hartree–Fock Theory

SIGERU HUZINAGA,
DENIS McWILLIAMS,*
and ANTONIO A. CANTU

*Department of Chemistry
The University of Alberta
Edmonton, Alberta, Canada*

I. Introduction	187
II. The Properties of Virtual Orbitals	189
A. Preliminaries	189
B. The Use of Projection Operators	192
C. The Choice of Ω	193
D. The Expansion Method	196
E. Discussion	199
III. Mathematical Background	199
IV. Theory of Separability of Many-Electron Systems	204
A. Concept of Separability	204
B. Setup of a Theoretical Model	205
C. Determination of the Optimum Φ_c	208
D. Hartree–Fock Approximation for Φ_c	211
E. Choice of “Core” Orbitals	213
F. The Diatomic Molecule	216
V. Concluding Remarks	217
Appendix	218
References	220

1. Introduction

Use of the projection operator technique in Hartree–Fock-type theories is so widespread and ubiquitous that it is beyond us to present a comprehensive review of the subject. We shall consequently only discuss some selected topics.

In the approximate treatment of many-electron systems, one often applies the variational procedure in which the wave function and accordingly the energy of the system are expressed as functionals of electron orbital functions $\{\phi_i\}$:

* Department of Chemistry, University of British Columbia, Vancouver, Canada.

$$\Phi \equiv \Phi(\{\phi_i\}), \quad (1.1)$$

$$E \equiv E(\{\phi_i\}). \quad (1.2)$$

In most cases the orthonormality conditions are imposed on $\{\phi_i\}$, namely,

$$\langle \phi_i | \phi_j \rangle = \langle \phi_j | \phi_i \rangle = \delta_{ij}. \quad (1.3)$$

These constraints may be incorporated into the variational treatment of the energy E by the use of Lagrangian multipliers. The resulting Euler equations may assume the form

$$\sum_k A_{ik} |\phi_k\rangle = \sum_j |\phi_j\rangle \theta_{ji}, \quad (1.4)$$

where the operators A_{ik} are functionals of $\{\phi_i\}$ and θ_{ji} are the Lagrangian multipliers. Equations (1.4) are of a very general form and constitute an adequate basis for a general multiconfiguration SCF theory (Huzinaga, 1969a). In the present article we shall be concerned with the case in which $A_{ik} = \delta_{ik} F_i$ and consequently Eqs. (1.4) reduce to

$$F_i |\phi_i\rangle = \sum_j |\phi_j\rangle \theta_{ji}, \quad (1.5)$$

where F_i are functionals of $\{\phi_i\}$. If the total wave function Φ and also F_i are both invariant with respect to arbitrary unitary transformation among $\{\phi_i\}$, one can eliminate the off-diagonal multipliers $\{\theta_{ji}\}$ ($j \neq i$) immediately. This is the case for a closed-shell system, but in the open-shell case it is not generally possible to utilize allowed unitary transformations among $\{\phi_i\}$ to eliminate the nondiagonal multipliers completely. There have been various proposals to deal with this problem. Among them one of the most successful so far is the coupling operator method proposed by Roothaan (1960). Mathematically speaking, it is a simple example of projection operator techniques.

By the use of appropriate coupling operators (Huzinaga, 1969b), one may bring the Hartree-Fock equation to the following form, including general open-shell cases:

$$R |\phi_i\rangle = |\phi_i\rangle \varepsilon_i, \quad (1.6)$$

where R is a functional of the occupied orbitals $\{\phi_i\}$. In solving Eq. (1.6) self-consistently, we pursue the establishment of a certain manifold, usually called the Hartree-Fock manifold, which consists of a finite set of functions to be identified as occupied orbitals. If there are N distinct occupied orbit-

als, the Hartree-Fock manifold is made of $\{\phi_i\}$, $i = 1, \dots, N$. In fact, this particular function space is the only theater which has physically solid meanings. However, Eq. (1.6) as an eigenvalue equation defines a complete set of functions and the Hartree-Fock manifold spans only a small subspace of the total function space. Sometimes, people wish to use eigenfunctions outside of the Hartree-Fock manifold, which are vaguely called virtual orbitals. It appears useful to manipulate artificially the eigenvalue spectrum of the virtual orbitals in certain meaningful ways and this may be done by adding terms which are essentially projection operators to the original Hartree-Fock operator. The subject is discussed in the second section from a practical point of view and in the third section some relevant mathematical background is described at some length.

A wide class of problems in atomic and molecular physics can be reduced to finding the extremum of the functional,

$$\langle \Psi | \mathcal{H} | \Psi \rangle / \langle \Psi | \Psi \rangle, \quad (1.7)$$

where $\mathcal{H}$ is a Hermitian operator and Ψ satisfies orthonormality constraints of the form

$$\langle \Psi | \phi_c \rangle = 0 \quad (1.8)$$

for some set of functions $\{\phi_c\}$ lying within a given subspace of the total function space. The set $\{\phi_c\}$ may span only a part (core region) of the Hartree-Fock manifold and Ψ occupies the rest—or Ψ could be a certain correlation function which is one-electron orthogonal to a Hartree-Fock “sea” represented by $\{\phi_c\}$. The generalized problem of variation in a restricted subspace is not an easy subject (Löwdin, 1965). In the fourth section of the present article we shall present a restricted discussion of this subject. Here again an extensive use of projection operator techniques is made.

II. The Properties of Virtual Orbitals

A. Preliminaries

In recent years, outstanding advances have been made in the design of electronic computers and this has made possible the large number of Hartree-Fock calculations that have appeared in the literature. Unfortunately, these calculations all display one rather unsatisfactory property. That is, the virtual orbital “energies” are usually positive, suggesting that they correspond to a state in the continuum.

In order to understand the origin of this defect we must return to the

method used to derive the Hartree-Fock equations themselves. For simplicity we will consider the closed-shell electron case, where the energy of the system is given by

$$E = 2 \sum_i^N H_i + \sum_{i \neq j}^N (2J_{ij} - K_{ij}) + \sum_i^N J_{ii}. \quad (2.1)$$

The variational principle may be applied to this expression, resulting in a set of self-consistent field equations,

$$F_i |\psi_i\rangle = \varepsilon_i |\psi_i\rangle, \quad (2.2)$$

where

$$F_i = -\frac{1}{2}\Delta - \sum_a (Z_a/r_a) + J_i + \sum_{i \neq j} (2J_j - K_j), \quad (2.3)$$

and J_j and K_j are the usual Coulomb and exchange operators.

However, this is not the usual procedure. It is customary to employ the relation,

$$J_{ii} - K_{ii} = 0 \quad (2.4)$$

so that the energy expression (2.1) may be rewritten as

$$E = 2 \sum_i^N H_i + \sum_{ij}^N (2J_{ij} - K_{ij}). \quad (2.5)$$

Applying the variational principle to Eq. (2.5) we obtain the well-known Hartree-Fock equations,

$$F |\psi_i\rangle = \varepsilon_i |\psi_i\rangle, \quad (2.6)$$

where

$$F = -\frac{1}{2}\Delta - \sum_a (Z_a/r_a) + \sum_j (2J_j - K_j). \quad (2.7)$$

Obviously, it is more convenient to use Eqs. (2.6) rather than Eqs. (2.3) since they have only a single F operator common to all the orbitals. Further, if $|\psi_i\rangle$ is an *occupied* orbital, Eqs. (2.3) and (2.6) are equivalent due to the relation,

$$(J_i - K_i) |\psi_i\rangle = 0. \quad (2.8)$$

However, they are not equivalent for the virtual orbitals. To understand this difference between the F_i operators and the Hartree-Fock operator, F , we must consider the potentials acting on an electron in an occupied

orbital, say $|\psi_i\rangle$. It will feel the influence of an operator J_j from each electron of opposite spin to itself and $(J_j - K_j)$ from each electron of the same spin. It will not, of course, be influenced by its own electronic charge, but it is just this effect that has been introduced into the F operator by using Eq. (2.4). Fortunately, Eq. (2.8) provides an "automatic" correction for the occupied orbitals.

The virtual orbitals undergo no such correction and the situation is represented by Fig. 1. The consequences are clear. If we use either F or F_i

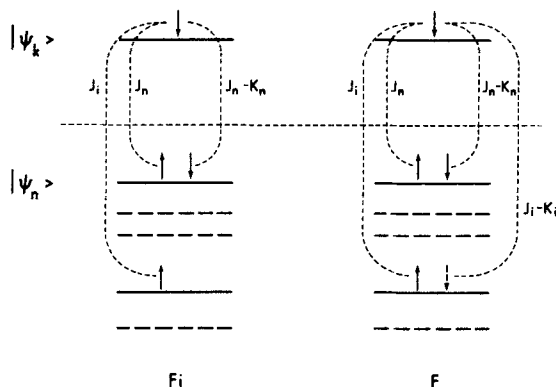


Fig. 1. Difference between F_i and F .

an electron in an occupied orbital feels the influence of the other $(2N - 1)$ electrons while one in a virtual orbital of F feels the full influence of $2N$ "core" electrons. On the other hand, an electron in a virtual orbital of F_i sees the core with an electron removed from the "occupied" $|\psi_i\rangle$. Thus the F_i produce virtual orbitals which correspond to the system under investigation, while F produces virtual orbitals more appropriate to the $(2N + 1)$ -electron system.

For many purposes, this distinction is not important, since the virtual orbitals may be of little interest. However, if they are to be used in, say, a perturbation expansion, then the effect could be significant. In such cases it would therefore seem reasonable to generate the virtual orbitals using one of the F_i operators.

This has been done successfully by Kelly (1964) in his perturbational calculations on beryllium. In his first paper (1963) he employed the Hartree-Fock operator and found that the virtual levels did not correspond to bound states. He was therefore able to use only continuum states in the

perturbation expansion, and convergence was slow. In his second paper (1964), he employed F_{2s} to generate the virtual levels. When these were included in the expansion, convergence was greatly improved, but they had one very undesirable property. They were not rigorously orthogonal to the occupied $1s$ orbital. This is an obvious consequence of the fact that $|1s\rangle$ is not an eigenfunction of F_{2s} , although it is one of both F and F_{1s} .

B. The Use of Projection Operators

The work of Kelly emphasized the need for a technique which would retain the physical principles mentioned above but which would avoid the nonorthogonality problem. This may be achieved by a straightforward mathematical device first discovered by Silverstone and Yin (1968). A simplified version of their method was proposed by Huzinaga and Arnau (1970, 1971).

Suppose that we have any Hermitian operator, R , and the associated eigenvalue problem,

$$R|\phi_i\rangle = |\phi_i\rangle\epsilon_i. \quad (2.9)$$

Let us now divide the eigenfunctions, $|\phi_i\rangle$, into "occupied" and "virtual" groups and define the operator,

$$V = (1 - P)\Omega(1 - P). \quad (2.10)$$

Here P is the projection operator,

$$P = \sum_i^{\text{occ}} |\phi_i\rangle\langle\phi_i| \quad (2.11)$$

and Ω is any arbitrary Hermitian operator.

Now if we define R' by

$$R' = R + V, \quad (2.12)$$

it is easy to see that for an occupied orbital,

$$R'|\phi_i\rangle = R|\phi_i\rangle = |\phi_i\rangle\epsilon_i, \quad (2.13)$$

while for a virtual orbital we have

$$R'|\phi_a\rangle = (R + \Omega - P\Omega)|\phi_a\rangle = |\phi_a\rangle\epsilon_a. \quad (2.14)$$

Of course, the ϵ_a and $|\phi_a\rangle$ of Eq. (2.14) are not the same virtual orbitals or energies as are obtained from (2.9) but have been modified by the presence of V . The occupied orbitals and energies, on the other hand, are unchanged but they are rigorously orthogonal to the new virtual orbitals because they are generated by a single Hermitian operator R' .

Further from Eq. (2-14) we have

$$\varepsilon_a = \langle \phi_a | R + \Omega | \phi_a \rangle, \quad (2.15)$$

which clearly indicates the relationship of the new energies to the R and Ω operators.

Obviously this technique can be used to modify the virtual orbitals of the Hartree-Fock operator and comparison of F_i and F indicates that appropriate choice of Ω would be

$$\Omega_i = -J_i + K_i. \quad (2.16)$$

Thus we obtain the modified equation,

$$[F - J_i + K_i + P(J_i - K_i)]|\psi_k\rangle = |\psi_k\rangle \varepsilon_k. \quad (2.17)$$

The virtual orbitals obtained from (2.17) will be almost, but not quite, identical to those obtained from F_i , and they will be rigorously orthogonal to all the occupied orbitals.

The virtual orbitals may be altered in an arbitrary manner since the only restriction placed on Ω is that it should be Hermitian. However, for a physically meaningful result we must select Ω with care, but even so there are many reasonable choices.

One important point must be stressed. The modified Hartree-Fock operator of Eq. (2.17) is not a straightforward addition of the F and Ω operators. Instead Ω has been guarded by the projection operator as shown in Eq. (2.10), so that there is no disturbance of the original occupied orbitals.

Another, practical, point to be mentioned is that we do not have to do any iterative SCF calculation with the modified Hartree-Fock operator. If the solution to the usual Hartree-Fock equations is known, all that is required is the construction of the modified operator and the solution of (2.17). We may simplify the matter even one step further, as will be shown later [Eq. (3.20)].

C. The Choice of Ω

The brief discussion on an appropriate form for Ω in the previous section of necessity related to the F and F_i operators of Eqs. (2.2) and (2.6). However, these refer to a closed-shell ground state which can be represented by a single Slater determinant. The virtual orbitals obtained from these also refer to a single Slater determinant with $m_s = 0$. This is unsatisfactory since such a wave function is not an eigenfunction of the S^2 operator.

If we assume that an electron has been excited from $|\psi_i\rangle$ to $|\psi_k\rangle$ and that neither is a member of a degenerate set, appropriate singlet and triplet eigenfunctions are

$$^1\Phi_{i \rightarrow k} = A\{\cdots (1/\sqrt{2})(\psi_i \alpha \psi_k \beta - \psi_i \beta \psi_k \alpha) \cdots\} \quad (2.18)$$

and

$$^3\Phi_{i \rightarrow k} = A\{\cdots \begin{bmatrix} \psi_i \alpha \psi_k \alpha \\ (1/\sqrt{2})(\psi_i \alpha \psi_k \beta + \psi_i \beta \psi_k \alpha) \\ \psi_i \beta \psi_k \beta \end{bmatrix} \cdots\}. \quad (2.19)$$

The corresponding energy expression is

$$\begin{aligned} E(^1, ^3\Phi_{i \rightarrow k}) &= 2 \sum_j^{\text{occ}} H_j + \sum_{jl}^{\text{occ}} (2J_{jl} - K_{jl}) - H_i \\ &\quad - \sum_j^{\text{occ}} (2J_{ji} - K_{ji}) + H_k + \sum_j^{\text{occ}} (2J_{jk} - K_{jk}) \\ &\quad - (J_{ik} - K_{ik}) \pm K_{ik}, \end{aligned} \quad (2.20)$$

where the plus and minus signs of the last term refer to the singlet and triplet, respectively. Hunt and Goddard (1969) have applied the variational principle to this expression and have obtained an "excited-state" operator suitable for the virtual orbitals

$$\begin{aligned} ^1, ^3F_i^{\text{EX}} &= -\frac{1}{2}\Delta - \sum_a (Z_a/r_a) + \sum_j (2J_j - K_j) - (J_i - K_i) \pm K_i \\ &= F - (J_i - K_i) \pm K_i. \end{aligned} \quad (2.21)$$

Thus a very desirable choice for Ω would be

$$^1, ^3\Omega_i = -(J_i - K_i) \pm K_i, \quad (2.22)$$

where the suffix i indicates that the potential is appropriate to a state where an electron has been excited from $|\psi_i\rangle$.

Using Eqs. (2.15) and (2.22) we obtain the new excited-state orbital energies

$$\begin{aligned} \varepsilon_k^{\text{EX}} &= \langle \psi_k | F | \psi_k \rangle - (J_{ik} - K_{ik}) \pm K_{ik} \\ &= \varepsilon'_{kk} - (J_{ik} - K_{ik}) \pm K_{ik}. \end{aligned} \quad (2.23)$$

It should be noted that the ε'_{kk} of Eq. (2.23) has the same mathematical form as the ε_k of the Hartree-Fock equations, but that these two quantities

are not equal. This is because the $|\psi_k\rangle$ are slightly different for the two operators. Another consequence of this is that the ε'_{kl} are not diagonal.

According to Roothaan (1951), the energy required to excite an electron from $|\psi_j\rangle$ to $|\psi_k\rangle$ is given by

$$E(^{1,3}\Phi_{j \rightarrow k}) - E(^1\Phi_0) = \varepsilon_k - \varepsilon_j - (J_{jk} - K_{jk}) \pm K_{jk}. \quad (2.24)$$

Substituting (2.23) into (2.24) we obtain,

$$\begin{aligned} E^{\text{EX}}(^{1,3}\Phi_{j \rightarrow k}) - E(^1\Phi_0) \\ = \varepsilon_k^{\text{EX}} - \varepsilon_j - (J_{jk} - K_{jk}) \pm K_{jk} + (J_{ik} - K_{ik}) \mp K_{ik}. \end{aligned} \quad (2.25)$$

Hence, if the potential employed is appropriate to the excitation being examined (that is if $i = j$), Eq. (2.25) reduces to

$$E^{\text{EX}}(^{1,3}\Phi_{i \rightarrow k}) - E(^1\Phi_0) = \varepsilon_k^{\text{EX}} - \varepsilon_i. \quad (2.26)$$

Thus the new orbital energies have a very simple physical interpretation and are appropriate for use in discussions of excitation energies. This is a definite conceptual improvement since it is difficult to visualize the relationship between the excitation energies and the old orbital energies. Mulliken (1949) has called the latter "unacceptable" and has defined $\varepsilon_k - J_{ik} + K_{ik}$ as being more meaningful. This, of course, corresponds to the F_i operator and is the average of the singlet and triplet cases outlined above.

Another pleasing aspect of using $^{1,3}\Omega_i$ is that the excited-state virtual orbitals depend on the orbital from which the electron is excited and on the multiplicity of the state formed. Both of these are essential from a physical point of view and the extent of the effect is illustrated by the Pariser-Parr-Pople calculation on aniline given in Fig. 2.

A great number of other potentials are, of course, possible. For example, in the atomic case an average correction could be obtained by using $-1/r_a$ since this would approximately cancel out one of the $2N$ core electrons. Comparison with (2.22) shows that this would yield an averaged triplet case since $1/r_a \sim (\sum_i J_i/N)$. Similarly (2.22) could be generalized to

$$\Omega_i = aJ_i + bK_i, \quad (2.27)$$

where the constants a and b would be chosen to suit the problem at hand. As mentioned previously, $a = -b = -1$ would yield an average of the singlet and triplet cases. There is therefore a great deal of latitude in the choice of Ω and the only limitations are that they are physically reasonable

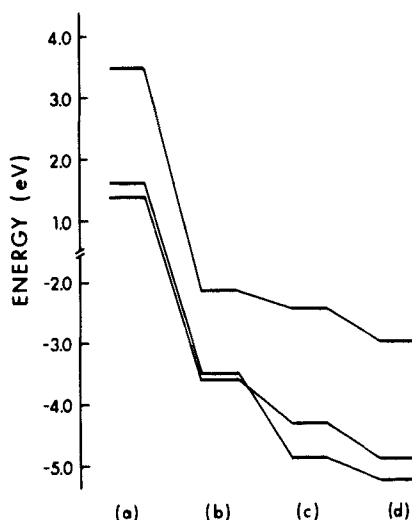


Fig. 2. The π virtual orbital energies obtained from a Pariser-Parr-Pople calculation on aniline using no modifying potential (a) and potentials appropriate for the following excitations: (b) highest occupied to a singlet state; (c) lowest occupied to a singlet state; (d) highest occupied to a triplet state.

and are suited to the level of approximation to be employed in the calculation.

D. The Expansion Method

In Roothaan's expansion method the $|\psi_i\rangle$ are expressed in terms of a finite basis set, $\{\chi_p\}$.

$$\psi_i = \sum_p c_{pi} \chi_p. \quad (2.28)$$

In this case the inclusion of the modifying potential follows closely that previously described. For simplicity we may assume that the $\{\chi_p\}$ and $\{c_{pi}\}$ are real and we define the matrices $\mathbf{D}$, $\mathbf{D}_j$, and $\mathbf{S}$ by

$$D_{pq} = \sum_j^{\text{occ}} c_{pj} c_{qj} = \sum_j^{\text{occ}} D_{j, pq}, \quad (2.29)$$

$$S_{pq} = \langle \chi_p | \chi_q \rangle. \quad (2.30)$$

The matrix corresponding to V in Eq. (2.10) is expressed as

$$\mathbf{V} = (\mathbf{I} - \mathbf{D})\mathbf{\Omega}(\mathbf{I} - \mathbf{D}), \quad (2.31)$$

where

$$\Omega_{pq} = \langle \chi_p | \Omega | \chi_q \rangle, \quad (2.32)$$

and $\mathbf{I}$ is the unit matrix. If we choose the Ω operator according to Eq. (2.27) the elements Ω_{pq} can easily be expressed in terms of two-electron integrals over the basis functions

$$\Omega_{i,pq} = \sum_{vw} \{a(pq|vw) + b(pw|vq)\} D_{i,vw}, \quad (2.33)$$

where

$$(pq|vw) = \int \chi_p(1) \chi_q(1) (1/r_{12}) \chi_v(2) \chi_w(2) dV_{12}. \quad (2.34)$$

All the equations of the previous sections have their matrix counterparts and we obtain an "excited-state" equation, which generates both occupied and virtual orbitals,

$$\mathbf{F}_i^{\text{EX}} \mathbf{C}_k^{\text{EX}} = \mathbf{S} \mathbf{C}_k^{\text{EX}} \epsilon_k, \quad (2.35)$$

where

$$\mathbf{F}_i^{\text{EX}} = \mathbf{F} + (\mathbf{I} - \mathbf{D}) \Omega_i (\mathbf{I} - \mathbf{D}), \quad (2.36)$$

and $\mathbf{F}$ is the usual Fock matrix.

In order to investigate the improvement we might expect that when using the expansion method we must consider the relationship between the old and new virtual orbitals. This can be described by the transformation

$$\mathbf{C}^{\text{EX}} = \mathbf{C} \mathbf{U}, \quad (2.37)$$

where $\mathbf{C}^{\text{EX}}$ is the $n \times q$ matrix with all the modified virtual orbital coefficients by columns, $\mathbf{C}$ is the corresponding matrix for the old virtual orbitals, and $\mathbf{U}$ is a $q \times q$ unitary matrix.

Let us now consider the sum of the virtual orbital energies:

$$\begin{aligned} \sum_k^{\text{VIRT}} \epsilon_k &= \text{Tr}(\mathbf{C}^+ \mathbf{F} \mathbf{C}) = \text{Tr}(\mathbf{U}^+ \mathbf{C}^+ \mathbf{F} \mathbf{C} \mathbf{U}) \\ &= \sum_k^{\text{VIRT}} \epsilon'_{kk}. \end{aligned} \quad (2.38)$$

Here ϵ'_{kk} is defined as in Eq. (2.23). Similarly, for the exchange and Coulomb integrals we obtain

$$\sum_k^{\text{VIRT}} K_{jk} = \sum_k^{\text{VIRT}} K_{jk}^{\text{EX}} \quad (2.39)$$

and

$$\sum_k^{\text{VIRT}} J_{jk} = \sum_k^{\text{VIRT}} J_{jk}^{\text{EX}}. \quad (2.40)$$

Summing Eqs. (2.24) and (2.25) over all the virtual orbitals and substituting Eqs. (2.38), (2.39), and (2.40) we have

$$\sum_k^{\text{VIRT}} E^{\text{EX}}(^m\Phi_{i \rightarrow k}) = \sum_k^{\text{VIRT}} E(^m\Phi_{i \rightarrow k}). \quad (2.41)$$

Thus, Eq. (2.41) shows that the energy of any given excited configuration can only be improved at the expense of at least one other. This is not necessarily a problem when a large basis can be employed, since many of the virtual orbitals generated will be of little interest. However, with a minimal basis set all the virtual orbitals are usually important and this restriction is serious.

Symmetry and orthogonality conditions also impose restrictions on minimal basis set calculations, as is clearly demonstrated by the virtual orbitals of pyridine (Fig. 3). The virtual orbital π_4 is entirely determined

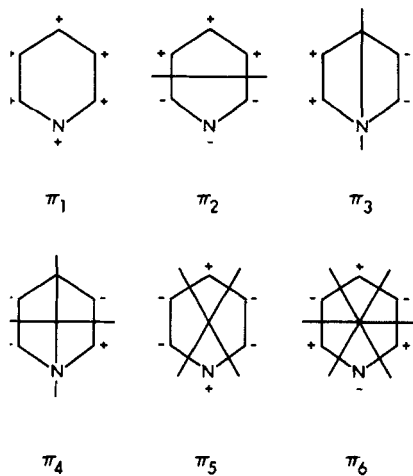


Fig. 3. The π -orbitals of pyridine.

by the occupied π_3 and cannot undergo any modification of the expansion coefficients. However, π_5 and π_6 can mix, allowing electron density to flow towards or away from the nitrogen atom. This is particularly important for $n \rightarrow \pi$ transitions since the lone-pair electrons are located mainly

on the nitrogen, and the use of ${}^1\Omega_n$ in a semiempirical all-valence-electron calculation resulted in the modified virtual orbitals

$$\pi_5^{\text{EX}} = 0.8909\pi_5 + 0.4543\pi_6, \quad (2.42)$$

$$\pi_6^{\text{EX}} = 0.4543\pi_5 - 0.8909\pi_6, \quad (2.43)$$

and an improvement in $E({}^1\Phi_{n \rightarrow 5})$ from -4199.33 eV to -4200.28 eV.

However, no overall improvement can be obtained with minimal basis sets. In fact, in some cases no modification at all is possible. Butadiene is such a case.

Further, it can easily be shown that Eq. (2.35) is entirely equivalent to a configuration interaction of the form

$${}^{1,3}\Psi_{ii} = \sum_k^{\text{VIRT}} {}^{1,3}\Phi_{i \rightarrow k} u_{ki}, \quad (2.44)$$

where the optimum values of the u_{ki} are the elements of the U matrix of Eq. (2.37). Therefore little improvement can be expected in a single excitation CI calculation, although there will be some change if double and higher excitations are included.

E. Discussion

As we have seen, when the expansion method is used with a minimal basis set, there is little room for maneuver. This is particularly true in semiempirical calculations where parameterization has already been employed to obtain the best possible result. Of course, this difficulty is not present in numerical Hartree-Fock calculations where we effectively have infinite freedom, and calculations with extended basis sets will lie between these two extremes.

Since the use of the potential (2.22) corresponds to improving the virtual orbitals by using a limited CI treatment, we may generally expect an improvement in the convergence of perturbation expansions. This is verified by the work of Kelly (1964) and the only application of the method at the time of writing is the work of Chase *et al.* (1971) on the correlation energy and Auger rates in neon.

III. Mathematical Background

Suppose we have an eigenvalue equation of an Hermitian operator H ,

$$H|\psi_i\rangle = \varepsilon_i|\psi_i\rangle. \quad (3.1)$$

For convenience, we divide the whole eigenspace into two subspaces, A and B :

$$\{|\psi_k\rangle\}, \quad k \in A \quad \text{and} \quad \{|\psi_m\rangle\}, \quad m \in B. \quad (3.2)$$

We also consider another eigenvalue equation,

$$(H + V)|\phi_i\rangle = \lambda_i|\phi_i\rangle, \quad (3.3)$$

where V is a projection of an arbitrary Hermitian operator Ω over one of the two subspaces, say A ,

$$V = P^+ \Omega P, \quad (3.4)$$

$$P = \sum_k^A |\psi_k\rangle \langle \psi_k|. \quad (3.5)$$

Using Eqs. (3.1), (3.4), and (3.5), we have

$$(H + V)|\psi_i\rangle = \varepsilon_i|\psi_i\rangle + \sum_{kl} \delta_{ik} |\psi_l\rangle \langle \psi_l | \Omega | \psi_k \rangle. \quad (3.6)$$

If $i \in B$,

$$(H + V)|\psi_i\rangle = \varepsilon_i|\psi_i\rangle, \quad (3.7)$$

which means

$$|\phi_i\rangle = |\psi_i\rangle, \quad \lambda_i = \varepsilon_i \quad \text{for } i \in B. \quad (3.8)$$

If $i \in A$,

$$(H + V)|\psi_i\rangle = \varepsilon_i|\psi_i\rangle + \sum_l^A |\psi_l\rangle \langle \psi_l | \Omega | \psi_i \rangle. \quad (3.9)$$

Suppose that A contains only one element, say k , then we have only one equation modified to be

$$(H + V)|\psi_k\rangle = (\varepsilon_k + \langle \psi_k | \Omega | \psi_k \rangle) |\psi_k\rangle,$$

which implies that all the eigenfunctions of $(H + V)$ remain the same as those of H but the particular k th eigenvalue is shifted by an amount of $\langle \psi_k | \Omega | \psi_k \rangle$. This property can be used in connection with deflation techniques in the diagonalization problem of secular equations. See, for instance, Wilkinson (1965) and Householder (1964). There is, however, an obvious possibility that the particular k th eigenvalue is shifted in such a way that it now coincides with one of other eigenvalues, say ε_m . This can be realized if

$$\langle \psi_k | \Omega | \psi_k \rangle = \varepsilon_m - \varepsilon_k. \quad (3.10)$$

Then the state m has an "accidental" degeneracy and an arbitrariness is introduced to the eigenfunctions: One may take any linear combination of $|\psi_k\rangle$ and $|\psi_m\rangle$. Furthermore, a similar but more general situation can arise even when A consists of two or more indices. In this case we may write Eq. (3.9) as

$$(H + V)|\psi_k\rangle = (\varepsilon_k + \langle\psi_k|\Omega|\psi_k\rangle)|\psi_k\rangle + \sum_{l \neq k}^A |\psi_l\rangle \langle\psi_l|\Omega|\psi_k\rangle. \quad (3.11)$$

Let us first suppose that the operator Ω is diagonal with respect to the eigenstates $\{|\psi_k\rangle\}$, $k \in A$;

$$\langle\psi_l|\Omega|\psi_k\rangle = b_k \delta_{lk}, \quad (3.12)$$

where b_k are any real numbers. Then we have

$$V = P^+ \Omega P = \sum_k^A b_k |\psi_k\rangle \langle\psi_k|, \quad (3.13)$$

and Eq. (3.11) may be written as

$$(H + V)|\psi_k\rangle = (\varepsilon_k + b_k)|\psi_k\rangle, \quad k \in A. \quad (3.14)$$

The implication becomes more apparent if we consider the case where all have a common shifting parameter, say ε_m ,

$$b_k = \varepsilon_m - \varepsilon_k. \quad (3.15)$$

Then the eigenvalue ε_m has now a manyfold degeneracy, and the eigenfunctions have accompanying arbitrariness. In the general case where the operator Ω is not necessarily diagonal, we define a new Hermitian matrix T by the elements

$$t_{lk} = \delta_{lk} \varepsilon_l + \langle\psi_l|\Omega|\psi_k\rangle, \quad k, l \in A. \quad (3.16)$$

Equation (3.11) can then be written as

$$(H + V)|\psi_k\rangle = \sum_l^A t_{lk} |\psi_l\rangle. \quad (3.17)$$

In Eq. (3.3) if $i \in B$ then Eqs. (3.8) are valid. If $i \in A$, Eq. (3.17) allows us to conclude that

$$|\phi_i\rangle = \sum_k^A c_{ki} |\psi_k\rangle, \quad i \in A. \quad (3.18)$$

Here the linear independence of $\{|\psi_k\rangle\}$ is utilized. The implication is that $\{|\phi_i\rangle\}$ ($i \in A$) are connected with $\{|\psi_i\rangle\}$ ($i \in A$) by a unitary matrix C

(c_{ki}). Substituting Eq. (3.18) into Eq. (3.3) and using the linear independence of $\{|\psi_i\rangle\}$ ($i \in A$), we are led to the relations

$$\sum_k t_{ik} c_{ki} = \lambda_i c_{ii}. \quad (3.19)$$

In a compact matrix form Eqs. (3.19) may be written

$$\mathbf{TC} = \mathbf{C}\mathbf{\Lambda}, \quad (3.20)$$

with the obvious definitions of matrices, $\mathbf{T}$, $\mathbf{C}$, and $\mathbf{\Lambda}$. The practical implication is that instead of solving the eigenvalue problem (3.3) to obtain the modified eigenvalues and eigenfunctions, the problem is reduced to the construction of an Hermitian matrix $\mathbf{T}$ and the diagonalization of the secular equation (3.20).

So far, we have tacitly had the virtual orbitals in mind and identified the subspace A with the virtual orbital space. At one point, however, we have come very close to the theory of the pseudopotential. In fact, it should be easily noticed that the projection operator V in Eq. (3.13) is a generalization of the pseudopotential introduced by Phillips and Kleinman (1959).

We do not intend to present a comprehensive general theory of the pseudopotential. Instead we shall present a brief sketch of it in relation to the use of projection operators.

Let us start again with Eqs. (3.1) and (3.3), but we do *not* specify the operator V from the outset and proceed along the line of Austin *et al.* (1962). As for the partition of the eigenspace we use $\{|\psi_c\rangle\}$ and $\{|\psi_v\rangle\}$ but this core-valence partition can be quite arbitrary from the mathematical point of view.

We assume that $|\phi_i\rangle$ can be expressed as

$$|\phi_i\rangle = \sum_c \alpha_c |\psi_c\rangle + \sum_v \alpha_v |\psi_v\rangle. \quad (3.21)$$

Then, from Eq. (3.3) we have

$$\begin{aligned} (H + V - \lambda_i)|\phi_i\rangle \\ = \sum_c \alpha_c (\epsilon_c - \lambda_i + V)|\psi_c\rangle + \sum_v \alpha_v (\epsilon_v - \lambda_i + V)|\psi_v\rangle = 0. \end{aligned} \quad (3.22)$$

Now let us characterize the operator V by specifying its projection properties.

Case 1 (Phillips-Kleinman)

$$\begin{aligned} V|\psi_c\rangle &= b_c|\psi_c\rangle \\ V|\psi_v\rangle &= 0 \end{aligned} \quad (3.23)$$

Here b_c is a number that can be viewed as an eigenvalue and V can be expressed as

$$V = \sum_c b_c |\psi_c\rangle \langle \psi_c|. \quad (3.24)$$

Equation (3.22) becomes

$$\sum_c \alpha_c (\varepsilon_c - \lambda_i + b_c) |\psi_c\rangle + \sum_v \alpha_v (\varepsilon_v - \lambda_i) |\psi_v\rangle = 0, \quad (3.25)$$

which means

$$\alpha_c (\varepsilon_c - \lambda_i + b_c) = 0, \quad (3.26)$$

$$\alpha_v (\varepsilon_v - \lambda_i) = 0. \quad (3.27)$$

If $(\varepsilon_v - \lambda_i) \neq 0$ for all v , then all the coefficients $\{\alpha_v\}$ vanish and at best we have

$$|\phi_i\rangle = \sum_c \alpha_c |\psi_c\rangle. \quad (3.28)$$

If λ_i coincides with one of $\{\varepsilon_v\}$, say $\varepsilon_v : (\varepsilon_v - \lambda_i) = 0$, then the particular α_v survives and we possibly have

$$|\phi_i\rangle = \sum_c \alpha_c |\psi_c\rangle + |\psi_v\rangle. \quad (3.29)$$

Now for $\{\alpha_c\}$ we have from Eq. (3.26)

$$\alpha_c (\varepsilon_c - \varepsilon_v + b_c) = 0. \quad (3.30)$$

Again, if $\varepsilon_c - \varepsilon_v + b_c \neq 0$ for all c , then all $\{\alpha_c\}$ should vanish and Eq. (3.29) gives simply $|\phi_i\rangle = |\psi_v\rangle$. Thus the necessary condition that we have Eq. (3.29) with nonvanishing $\{\alpha_c\}$ is that

$$\varepsilon_c - \varepsilon_v + b_c = 0. \quad (3.31)$$

Going back to Eq. (3.24), we have

$$V = \sum_c (\varepsilon_v - \varepsilon_c) |\psi_c\rangle \langle \psi_c|, \quad (3.32)$$

which is the Phillips-Kleinman pseudopotential.

Case 2

$$\begin{aligned} V|\psi_c\rangle &= \sum_{c'} b_{c',c} |\psi_{c'}\rangle \\ V|\psi_v\rangle &= 0 \end{aligned} \quad (3.33)$$

The above operator V can be Hermitian and be expressed as

$$V = \sum_{c',c} b_{c',c} |\psi_{c'}\rangle \langle \psi_c|. \quad (3.34)$$

It is now clear that we are talking about the operator which is mathematically the same as the one in Eq. (3.4.)

Case 3 (Austin–Heine–Sham)

$$\begin{aligned} V|\psi_c\rangle &= \sum_{c'} b_{c',c} |\psi_{c'}\rangle \\ V|\psi_v\rangle &= \sum_c d_{c,v} |\psi_c\rangle \end{aligned} \quad (3.35)$$

It is obvious that the operator V in Eqs. (3.35) cannot be Hermitian. In the projection operator form it may be written

$$V = \sum_c |\psi_c\rangle \langle F_c|. \quad (3.36)$$

Here, the F_c are completely arbitrary functions and this is the Austin–Heine–Sham pseudopotential (1962).

Case 4

$$\begin{aligned} V|\psi_c\rangle &= \sum_{c'} b_{c',c} |\psi_{c'}\rangle + \sum_v e_{v,c} |\psi_v\rangle \\ V|\psi_v\rangle &= \sum_c d_{c,v} |\psi_c\rangle + \sum_{v'} g_{v',v} |\psi_{v'}\rangle \end{aligned} \quad (3.37)$$

This is the most general case in which the operator V can be Hermitian with additional restrictions among the coefficients. It presents a theoretical basis for the discussion of various model potentials.

IV. Theory of Separability of Many-Electron Systems

A. Concept of Separability

The idea of separating the electrons of an atomic or molecular system into sets describing “relevant” regions (or groups) and sets describing “unimportant” regions has indeed been among chemists for some decades. Examples of such an idea and some of the terminology that has evolved from this concept include the “core” and “valence” electrons of atoms, the σ and π electrons of conjugated hydrocarbons, functional groups of organic chemistry, and “valence” and “core” regions in spatially separated interacting systems. The latter example, of course, includes the previous examples as special cases as well as a polyatomic molecule and a stage of a chemical reaction involving several systems. The case of just two interacting systems, which we shall hereafter consider,

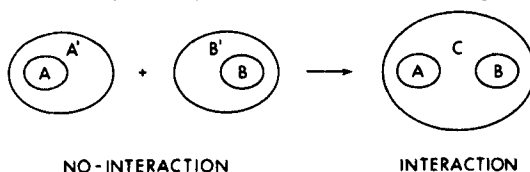


Fig. 4 Two interacting systems. Regions A and B are taken as “unimportant” and noninteracting. For R being any of the regions we let N_R be its number of electrons. The interacting system is taken to have $N = N_A + N_B + N_C$ electrons where $N_C = N_{A'} + N_{B'}$.

exemplifies the separability idea quite generally and this situation is schematically illustrated in Fig. 4.

Intuitively the meaning of “relevant” and “unimportant” regions is rather naive. With regards to our example we can say that the A and B parts are “unimportant” and are thus unaffected by the chemical behavior of the (chemically) “relevant” part C . This can be taken to mean that the states and energies of the total system are governed by the states and energies of the “relevant” part.

Our goal shall be to provide a theoretical meaning or model for the separability idea. Any such model, whose ultimate purpose should be to properly provide states and energies of the “relevant” region C , will involve a determination of an effective-Hamiltonian for the C region which somehow incorporates the effect of the A and B parts, as is the case, for example, when one uses pseudo or model potentials. We shall, however, avoid the commonly used notion of pseudopotentials and develop an effective-Hamiltonian which accounts for the A and B effects through appropriate *projection operators*. In this way, for example, we can assure that the molecular nodal characteristics of the system wave function are not disturbed. Along these lines, Szasz (1959) proceeded to develop such a model based on projection operators, and some errors in his formalism were pointed out by Weeks and Rice (1968). We used these developments together with ideas developed by McWeeny (1959) to provide a theory which leads to possible new simplifications in the treatment of complex many-electron systems.

B. Setup of a Theoretical Model

This part is founded on the works of McWeeny (1959) and Lykos and Parr (1956). Credit is also due to the early work of Fock *et al.* (1940) as well as the work of Szász (1959). Hence we shall only briefly present the ideas of these workers in a form relevant to our problem.

An antisymmetric and normalized wave function¹ of the N -electron interacting system of Fig. 4 can be expressed as

$$\Psi = \Psi_{ABC} = M \tilde{A}_p(\Phi_A \Phi_B \Phi_C), \quad (4.1)$$

where each of the "group" functions,

$$\begin{aligned} \Phi_A &= \Phi_A(1, 2, \dots, N_A), \\ \Phi_B &= \Phi_B(N_A + 1, N_A + 2, \dots, N_A + N_B), \\ \Phi_C &= \Phi_C(N_A + N_B + 1, \dots, N_A + N_B + N_C) \end{aligned}$$

is antisymmetric and normalized, M is a normalization constant, and $\tilde{A}_p$ is the partial antisymmetrizer. It should be noted that each Φ_R can be a linear combination of some product-type function as mentioned by Lykos and Parr (1956),

$$\Phi_R = R_1 \Phi_{R,1} + R_2 \Phi_{R,2} + \dots \quad (4.2)$$

We shall assume that each total Ψ satisfies the A - B - C separability condition. This merely means that Φ_A , Φ_B , and Φ_C are strongly orthogonal, i.e., for $R \neq S \in \{A, B, C\}$,

$$\int \Phi_R^*(1, i, j, \dots) \Phi_S(1, k, m, \dots) d\tau_1 = 0. \quad (4.3)$$

In accordance with the partitioning of Ψ in Eq. (4.1), the spin-free electronic Hamiltonian can be written as

$$\mathcal{H} = (\mathcal{H}^A + \mathcal{H}^B + \mathcal{H}^{A,B}) + (\mathcal{H}^C + \mathcal{H}^{A,C} + \mathcal{H}^{B,C}), \quad (4.4)$$

where, for groups $R \neq S \in \{A, B, C\}$,

$$\mathcal{H}^{R,S} = \sum_{\substack{\nu \in R \\ \mu \in S}} 1/r_{\mu\nu} \quad (4.5)$$

is the electron repulsion term between the electrons in group R and group S , and

$$\mathcal{H}^R = \sum_{\mu \in R} h(\mu) + \sum_{\substack{\mu < \nu \\ \mu, \nu \in R}} 1/r_{\mu\nu}, \quad (4.6)$$

with

$$h(\mu) = -\frac{1}{2}\Delta_\mu + V(\mu), \quad (4.7)$$

¹A wave function is taken to mean an approximate state. We assume at the outset that we have adequate wave functions.

where $V(\mu)$ is the potential energy of electron μ in the field of the nuclei of the total system. Using this form of $\mathcal{H}$, the total energy expression of the system for a state approximated by an A - B - C separable wave function becomes

$$E = \langle \Psi | \mathcal{H} | \Psi \rangle = H^{A,B} + \hat{H}^C. \quad (4.8)$$

Here $H^{A,B}$ describes the energy of the A, B "core";

$$H^{A,B} = H^A + H^B + J^{AB} - K^{AB}, \quad (4.9)$$

where, for $R \neq S \in \{A, B\}$,

$$H^R = \langle \Phi_R | \mathcal{H}^R | \Phi_R \rangle, \quad (4.10)$$

$$J^{RS} = \int (1/r_{12}) P_1^R(1; 1) P_1^S(2; 2) dV_1 dV_2, \quad (4.11)$$

$$K^{RS} = \frac{1}{2} \int (1/r_{12}) P_1^R(2; 1) P_1^S(1; 2) dV_1 dV_2, \quad (4.12)$$

and P_1^R , for example, is the spinless one-particle density matrix for Φ_R . $\hat{H}^C$, on the other hand, describes the energy of the "important" region C ;

$$\hat{H}^C = \langle \Phi_C | \mathcal{H}^C | \Phi_C \rangle \quad (4.13)$$

where

$$\mathcal{H}^C = \sum_{\mu \in C} \hat{h}(\mu) + \sum_{\substack{\mu < \nu \\ \mu, \nu \in C}} 1/r_{\mu\nu} \quad (4.14)$$

and

$$\hat{h}(\mu) = h(\mu) + \mathcal{J}^A(\mu) - \mathcal{K}^A(\mu) + \mathcal{J}^B(\mu) - \mathcal{K}^B(\mu). \quad (4.15)$$

Here $\mathcal{J}^A(\mu)$ and $\mathcal{K}^A(\mu)$, for example, are appropriately defined Coulomb and exchange integral operators, respectively.

The energy expression (4.8) is a generalization of the Fock, Wesselov, and Petraschen (1940) formula. Furthermore, in the form written, it takes the "core" form of the "core"-peel relation of Lykos and Parr (1956). We should note that $\mathcal{H}^C$ operates only on the C electrons and has the effect of the A, B "core" incorporated into it through $\mathcal{J}^R(\mu)$ and $\mathcal{K}^R(\mu)$, $R \in \{A, B\}$. We shall assume now that the system satisfies the C -electron approximation, that is, there exists a set of states which are adequately approximated by a set of wave functions $\{\Psi_{ABC}\}$ each of which is A - B - C

separable and, for the set, Φ_A and Φ_B are taken as fixed. One can equivalently say that we are neglecting the effect of the C electrons on the A and B electrons, or, more crudely, the behavior of the system is determined by the states of the C region.

C. Determination of the Optimum Φ_C

We shall hereafter assume that the system satisfies the C -electron approximation. Consequently the equation for the optimum Φ_C will serve for a set of approximate states of the system. Following Fock *et al.* (1940), let us take the further approximation that each of the "core" Φ_A and Φ_B is an antisymmetrized product of orthonormal spin-orbitals,

$$\Phi_R = \frac{1}{\sqrt{N_R!}} \det[\phi_1^R \phi_2^R \dots \phi_{N_R}^R] \quad (4.16)$$

and

$$\langle \phi_i^R | \phi_j^R \rangle = \delta_{ij} \quad \text{for } i, j = 1, \dots, N_R, \quad (4.17)$$

where $R \in \{A, B\}$. The strong orthogonality among Φ^A , Φ^B , and Φ^C then becomes

$$\int \Phi_C^*(1, i, j, \dots) \phi_p^R(1) d\tau_1 = 0 \quad \text{for } \begin{matrix} R \in \{A, B\} \\ p = 1, \dots, N_R \end{matrix} \quad (4.18)$$

and

$$\langle \phi_i^A | \phi_j^B \rangle = 0 \quad \text{for } \begin{matrix} i = 1, \dots, N_A \\ j = 1, \dots, N_B \end{matrix}. \quad (4.19)$$

Regardless of the choice of spin-orbitals $\{\phi_i^R\}$, $R \in \{A, B\}$, we will always have an upper bound to the total energies of the system. Assuming that we hold $\{\phi_i^R\}$, $R \in \{A, B\}$, fixed in some convenient and physically meaningful way, we shall derive the equation for the optimum Φ_C . This was essentially done by Weeks and Rice (1968) but their main concern was the derivation of a formal pseudopotential type equation. Our emphasis, as stated before, is different from theirs in that we rather deliberately avoid the pseudopotential-pseudo-wave-function approach.

Let us begin by defining the projection operator Q as

$$(1 - Q) = (1 - Q^A)(1 - Q^B), \quad (4.20)$$

where, for $R \in \{A, B\}$,

$$(1 - Q^R) = \prod_{\mu \in C} (1 - P_\mu^R) \quad (4.21)$$

and

$$P_{\mu}^R = \sum_{k=1}^{N_R} |\phi_k^R(\mu)\rangle \langle \phi_k^R(\mu)|. \quad (4.22)$$

Some properties of Q are

$$(a) \quad (1 - Q) = (1 - Q^A)(1 - Q^B) = \prod_{\mu \in C} (1 - P_{\mu}^A - P_{\mu}^B); \quad (4.23)$$

$$(b) \quad Q^R(1 - Q) = (1 - Q)Q^R = 0 \quad \text{for} \quad R \in \{A, B\}; \quad (4.24)$$

$$(c) \quad Q(1 - Q) = (1 - Q)Q = 0; \quad (4.25)$$

$$(d) \quad |\Phi\rangle \text{ is strongly orthogonal to } |\Phi_A\rangle \text{ and } |\Phi_B\rangle \text{ if and only if}$$

$$Q|\Phi\rangle = 0. \quad (4.26)$$

For a nonzero $|\Phi_C\rangle$ that satisfies Eq. (4.26), there exists a $|\Phi_C^{(0)}\rangle$ such that

$$|\Phi_C\rangle = (1 - Q)|\Phi_C^{(0)}\rangle. \quad (4.27)$$

This $|\Phi_C^{(0)}\rangle$ is free from the requirement of satisfying Eq. (4.26).

In the energy expression (4.8) let us take the "core" part $H^{A,B}$ as fixed and treat

$$\hat{H}^C = \langle \Phi_C | \mathcal{H}^C | \Phi_C \rangle = \langle \Phi_C^{(0)} | (1 - Q) \mathcal{H}^C (1 - Q) | \Phi_C^{(0)} \rangle \quad (4.28)$$

variationally with the normalization constraint

$$\langle \Phi_C | \Phi_C \rangle = \langle \Phi_C^{(0)} | (1 - Q) | \Phi_C^{(0)} \rangle = 1. \quad (4.29)$$

The resulting Euler equation reduces to

$$[(1 - Q) \mathcal{H}^C (1 - Q)] |\Phi_C^{(0)}\rangle = E_C (1 - Q) |\Phi_C^{(0)}\rangle \quad (4.30)$$

or, by using (4.27), to

$$(1 - Q) \mathcal{H}^C |\Phi_C\rangle = E_C |\Phi_C\rangle, \quad (4.31)$$

where E_C is the Lagrangian multiplier. Note that Q , $\mathcal{H}^C$, and consequently $(1 - Q) \mathcal{H}^C$ do not depend on $|\Phi_C\rangle$. Since $(1 - Q) \mathcal{H}^C$ is not Hermitian unless $[\mathcal{H}^C, Q] = 0$ and since $|\Phi_C\rangle$ must satisfy (4.26), then $(1 - Q) \mathcal{H}^C$ cannot be used to determine upper bounds to E_C variationally. However, the Hermitian operator (Huzinaga and Cantu, 1971),

$$\mathcal{H}^C = (1 - Q) \mathcal{H}^C - \mathcal{H}^C Q \quad (4.32)$$

possesses an energy spectrum that overlaps with set of energy values E_C from (4.30). Hence $\mathcal{H}^C$ is a proper Hermitian effective Hamiltonian which

can be used in a straightforward variational calculation to determine upper bounds to the lowest E_C , if the term $-\mathcal{H}^C Q$ does not create eigenvalues lower than the lowest in Eq. (4.31). This condition will be met in most cases of real interest.

If we can divide $\mathcal{H}^C$ as

$$\mathcal{H}^C = \mathcal{H}_\alpha^C + \mathcal{H}_\beta^C \quad (4.33)$$

where

$$[\mathcal{H}_\alpha^C, Q] = 0 \quad \text{and} \quad [\mathcal{H}_\beta^C, Q] \neq 0 \quad (4.34)$$

then

$$\mathcal{H}'^C = (1 - Q)\mathcal{H}^C - \mathcal{H}_\beta^C Q \quad (4.35)$$

will still serve as a proper effective Hamiltonian. Szász (1959) proposed as a proper effective Hamiltonian the operator

$$\mathcal{H}_{SZ}^C = \mathcal{H}^C - Q\mathcal{H}_\beta^C, \quad (4.36)$$

which gives the same expectation value as that of Eq. (4.35) provided Eq. (4.26) is used. However, this $\mathcal{H}_{SZ}^C$ will have eigenvalues lower than the lowest E_C in Eq. (4.31). Consequently, its use will lead to the determination of wrong energy values.

$\mathcal{H}^C$ or $\mathcal{H}'^C$ of Eq. (4.32) and (4.35), respectively, is in a general form but a drawback is that Q is quite an involved nonlocal operator when the number of electrons is large. If Φ_C takes the form [see Eq. (4.2)]

$$\Phi_C = \sum_i C_i \Phi_{C,i}, \quad (4.37)$$

where each $\Phi_{C,i}$ is a Slater determinant built from a set of orthonormal spin-orbitals, then, as shown in the Appendix, $\mathcal{H}^C$ (or $\mathcal{H}'^C$) has an equivalent simplified form given by

$$\begin{aligned} \mathcal{H}^C \approx \mathcal{H}_{\text{equiv}}^C &= \sum_{\mu \in C} [(1 - P_\mu)\hat{h}(\mu) - \hat{h}(\mu)P_\mu] \\ &+ \sum_{\substack{\nu < \mu \\ \mu, \nu \in C}} \left[(1 - P_\mu)(1 - P_\nu) \frac{1}{r_{\mu\nu}} + \frac{1}{r_{\mu\nu}} (P_\mu P_\nu - P_\mu - P_\nu) \right], \end{aligned} \quad (4.38)$$

where P_μ is equal to $P_\mu^A + P_\mu^B$ and the equivalence, $\approx$, means that both Hamiltonians give the same energy expression upon using Φ_C of Eq. (4.37).

D. Hartree-Fock Approximation for Φ_C

We shall consider the situation where all three, Φ_A , Φ_B , and Φ_C , are doubly occupied, closed-shell singlet functions. Each of these can then be written as

$$\Phi_R = \frac{1}{\sqrt{N_R!}} A[|\psi_1^R \psi_1^R \psi_2^R \psi_2^R \dots \psi_{N_R/2}^R \psi_{N_R/2}^R\rangle |\alpha\beta\alpha\beta \dots \alpha\beta\rangle] \quad (4.39)$$

with

$$\langle \psi_i^R | \psi_j^R \rangle = \delta_{ij} \quad \text{for} \quad i, j = 1, \dots, N_R/2, \quad (4.40)$$

where $R \in \{A, B, C\}$, A is the antisymmetrizer, and the first and second ket after the antisymmetrizer are the spatial and spin N_R -electron kets, respectively. As before, let us hold Φ_A and Φ_B fixed and determine the optimum Φ_C , or $\{\psi_i^C\}$ in this case, under the strong orthogonality constraints which reduce to

$$\langle \psi_i^A | \psi_j^B \rangle = \langle \psi_i^A | \psi_k^C \rangle = \langle \psi_j^B | \psi_k^C \rangle = 0, \quad (4.41)$$

with

$$\begin{aligned} i &= 1, \dots, N_A/2 \\ j &= 1, \dots, N_B/2 \\ k &= 1, \dots, N_C/2 \end{aligned}$$

The energy expression in (4.13) becomes

$$\begin{aligned} \hat{H}^C &= \langle \Phi_C | \hat{\mathcal{H}}^C | \Phi_C \rangle \\ &= \sum_{i=1}^{N_C/2} \langle \psi_i^C | \hat{h} | \psi_i^C \rangle + \sum_{i,j=1}^{N_C/2} (2J_{ij}^C - K_{ij}^C), \end{aligned} \quad (4.42)$$

where [see Eq. (4.15)]

$$\hat{h} = h + \sum_{j=1}^{N_A/2} (2J_j^A - K_j^A) + \sum_{j=1}^{N_B/2} (2J_j^B - K_j^B). \quad (4.43)$$

Here the numbers J_{ij}^C and K_{ij}^C are Coulomb and exchange integrals, respectively, defined over the spatial orbitals and J_i^R and K_i^R are Coulomb and exchange integral operators, respectively, defined also over the spatial orbitals. Upon proceeding with our variational treatment we arrive at the Euler equation

$$F|\psi_i^C\rangle = \sum_{j=1}^{N_C/2} \varepsilon_{ij}^C |\psi_j^C\rangle + \sum_{k=1}^{N_A/2} \lambda_{ik}^A |\psi_k^A\rangle + \sum_{l=1}^{N_B/2} \lambda_{il}^B |\psi_l^B\rangle, \quad (4.44)$$

where

$$F = \hat{h} + \sum_{k=1}^{N_C/2} (2J_j^C - K_j^C). \quad (4.45)$$

The Lagrangian multipliers $\{\varepsilon_{ij}^C\}$ can be diagonalized as usual by a unitary transformation while the remaining multipliers, $\{\lambda_{ij}^A\}$ and $\{\lambda_{ij}^B\}$, can be processed by the coupling operator technique (Huzinaga, 1969b). Thus Eq. (4.44) can be brought to the form

$$F|\psi_i^C\rangle = \varepsilon_i^C|\psi_i^C\rangle + P_A F|\psi_i^C\rangle + P_B F|\psi_i^C\rangle \quad (4.46)$$

or

$$[1 - (P_A + P_B)]F|\psi_i^C\rangle = \varepsilon_i^C|\psi_i^C\rangle, \quad (4.47)$$

where, for $R \in \{A, B\}$,

$$P_R = \sum_{i=1}^{N_R/2} |\psi_i^R\rangle\langle\psi_i^R|. \quad (4.48)$$

If we write down the full expression of F in Eq. (4.45) we obtain

$$F = -\frac{1}{2}\Delta + V + \sum_{i=1}^{N_A/2} (2J_i^A - K_i^A) + \sum_{i=1}^{N_B/2} (2J_i^B - K_i^B) + \sum_{i=1}^{N_C/2} (2J_i^C - K_i^C). \quad (4.49)$$

which is the Hartree-Fock operator that determines the best orbitals in the *total* wave function Ψ_{ABC} of (4.1) [with (4.39) as group orbitals].

Generally speaking we have

$$[(P_A + P_B), F] \neq 0, \quad (4.50)$$

in which case $[1 - (P_A + P_B)]F$ is not Hermitian and, from the previous section, the proper Hermitian Hartree-Fock operator is then

$$\tilde{F} = [1 - (P_A + P_B)]F - F(P_A + P_B). \quad (4.51)$$

However, when

$$[(P_A + P_B), F] = 0 \quad (4.52)$$

$\tilde{F}$ can simply be

$$\tilde{F} = [1 - (P_A + P_B)]F. \quad (4.53)$$

E. Choice of "Core" Orbitals

The question of commutability between $(P_A + P_B)$ and F , of course, depends on the choice of the fixed orbitals $\{\psi_i^A\}$ and $\{\psi_i^B\}$. We shall consider three possibilities. The first choice is the Szász (1959) choice,

$$\hat{h}^A |\psi_i^A\rangle = \omega_i^A |\psi_i^A\rangle \quad (4.54)$$

and

$$\hat{h}^B |\psi_i^B\rangle = \omega_i^B |\psi_i^B\rangle, \quad (4.55)$$

where

$$\hat{h}^R = -\frac{1}{2}\Delta + V^R + \sum_{i=1}^{N_R/2} (2J_i^R - K_i^R) \quad (4.56)$$

is the Hartree-Fock operator for the "core"- R orbitals, R being A or B . Here V^R is such that $V^R(\mu)$ is the potential energy of electron μ in the field of the nuclei in "core" R . Since

$$V = V^A + V^B + V^C, \quad (4.57)$$

we see that

$$F = \hat{h}^A + \left[V^B + \sum_{i=1}^{N_B/2} (2J_i^B - K_i^B) \right] + \left[V^C + \sum_{i=1}^{N_C/2} (2J_i^C - K_i^C) \right] \quad (4.58)$$

or

$$F = \hat{h}^B + \left[V^A + \sum_{i=1}^{N_A/2} (2J_i^A - K_i^A) \right] + \left[V^C + \sum_{i=1}^{N_C/2} (2J_i^C - K_i^C) \right]. \quad (4.59)$$

In order for F and $(P_A + P_B)$ to commute it is necessary, but not sufficient, for the last term of (4.58) or (4.59) to vanish upon applying $P_A + P_B$ to it. As this is never the case we conclude that with this first (Szász) choice, F and $(P_A + P_B)$ do not commute and $\tilde{F}$ must be written as (4.51).

Our second choice is simply

$$F |\psi_i^A\rangle = \lambda_i^A |\psi_i^A\rangle \quad \text{with} \quad \lambda_1^A < \cdots < \lambda_{N_A/2}^A \quad (4.60)$$

and

$$F |\psi_j^B\rangle = \lambda_j^B |\psi_j^B\rangle \quad \text{with} \quad \lambda_{N_A/2}^A < \lambda_j^B, \quad (4.61)$$

where, in Eq. (4.61), $j = 1, 2, \dots, N_B/2$. From Eq. (4.49), this choice amounts to choosing the "core" orbitals to be the lowest set of Hartree-Fock orbitals from the set of Hartree-Fock orbitals of the *entire* system. Clearly $(P_A + P_B)$ and F commute and thus $\tilde{F}$ can be written as Eq. (4.53), which becomes, by Eqs. (4.48), (4.60), and (4.61),

$$\tilde{F} = F - \sum_{i=1}^{N_A/2} \lambda_i^A |\psi_i^A\rangle\langle\psi_i^A| - \sum_{i=1}^{N_B/2} \lambda_i^B |\psi_i^B\rangle\langle\psi_i^B|. \quad (4.62)$$

This suggests a new alternative form for $\mathcal{H}^C$ of Eq. (4.32):

$$\mathcal{H}_{\text{ALT}}^C = \sum_{\mu \in C} \tilde{h}(\mu) + \sum_{\substack{\mu < \nu \\ \mu, \nu \in C}} 1/r_{\mu\nu}, \quad (4.63)$$

where

$$\tilde{h}(\mu) = \hat{h}(\mu) - \sum_{i=1}^{N_A/2} \lambda_i^A |\psi_i^A\rangle\langle\psi_i^A| - \sum_{i=1}^{N_B/2} \lambda_i^B |\psi_i^B\rangle\langle\psi_i^B| \quad (4.64)$$

and $\hat{h}(\mu)$ is given in (4.43) as $\hat{h}$. By alternative we mean that upon optimizing $\langle\Phi_C|\mathcal{H}_{\text{ALT}}^C|\Phi_C\rangle$ using Φ_C of (4.39) and *only* the orthogonormality constraints (4.40) within $\{\psi_i^C\}$, i.e., upon taking

$$\delta \left[\langle\Phi_C|\mathcal{H}_{\text{ALT}}^C|\Phi_C\rangle - \sum_{i,j=1}^{N_c} \varepsilon_{ij}^C \langle\psi_i^C|\psi_j^C\rangle \right] = 0, \quad (4.65)$$

we obtain

$$\tilde{F}|\psi_i^C\rangle = \varepsilon_i^C|\psi_i^C\rangle, \quad (4.66)$$

where $\tilde{F}$ is given in (4.62). One might be tempted to take the last two terms in (4.62) or (4.64) and interpret them as, or replace them by, some kind of local pseudopotential, as is done by Szász and McGinn (1966). As may be recalled, this is not our main concern.

The use of $\mathcal{H}_{\text{ALT}}^C$ is of some interest. Suppose that we use in $\langle\Phi_C|\mathcal{H}_{\text{ALT}}^C|\Phi_C\rangle$ a normalized Φ_C which is more versatile than the single Slater determinant in Eq. (4.39) [and its constraints (4.40)] as, for example, the configuration interaction form in (4.37). Then, by the variation principle, this versatility will yield for us, upon optimization, a *lower* energy than the Hartree-Fock energy. We can, however, not guarantee anything on the comparison of this energy with the ones obtained when $\mathcal{H}^C$ of (4.32) or $\mathcal{H}_{\text{equiv}}^C$ of (4.38) is used.

Our final choice for the “core” orbitals is an intermediate case:

$$F_A |\psi_i^A\rangle = \varepsilon_i^A |\psi_i^A\rangle \quad (4.67)$$

and

$$F_B |\psi_i^B\rangle = \varepsilon_i^B |\psi_i^B\rangle \quad (4.68)$$

where, for $R \in \{A, B\}$,

$$F_R = \hat{h}^R + \sum_{i=1}^{N_R/2} (2J_i^{R'} - K_i^{R'}) \quad (4.69)$$

is the Hartree-Fock operator of a subsystem having a “core” R and contributing N_R electrons to the region C (see Fig. 4.), and $\hat{h}^R$ is given in Eq. (4.56). Clearly, $N_C = N_{A'} + N_B$, as mentioned in Fig. 4. As done in (4.58), the Hartree-Fock operator F of (4.49) can be written, for this case, as

$$\begin{aligned} F = F_A + & \left[V^B + \sum_{i=1}^{N_B/2} (2J_i^B - K_i^B) \right] \\ & + \left[V^C + \sum_{i=1}^{N_C/2} (2J_i^C - K_i^C) - \sum_{i=1}^{N_{A'}/2} (2J_i^{A'} - K_i^{A'}) \right]. \end{aligned} \quad (4.70)$$

In parallel with (4.59), another form can be written where $A(A')$ and B are interchanged in (4.70). If the interacting system is sufficiently separated into its interacting components and $V^C = 0$ (as when C is a pure electron-bonding region) then

$$P_A \left\{ V^B + \sum_{i=1}^{N_B/2} (2J_i^B - K_i^B) \right\} \simeq 0 \quad (4.71)$$

and

$$P_A \left\{ V^C + \sum_{i=1}^{N_C/2} (2J_i^C - K_i^C) - \sum_{i=1}^{N_{A'}/2} (2J_i^{A'} - K_i^{A'}) \right\} \simeq 0 \quad (4.72)$$

with a similar result holding when $A(A')$ and B are interchanged. If indeed the system is such that the approximations (4.71) and (4.72) hold, then $(P_A + P_B)$ and F very nearly commute and thus this case effectively reduces to that of the second choice. Equations (4.62) to (4.66) would then follow, with λ_i^R replaced by ε_i^R where $R \in \{A, B\}$.

F. The Diatomic Molecule

Before proceeding to the diatomic molecule case, we mention that the atomic case with "core" and "valence" electrons or, for that matter, any case with only a "core" and "valence" region, is obtainable from the previous treatment by taking, say, $B = 0$. Then only the mentioned cases, 1 and 2, of Section IV.E hold for the "core" orbitals. Consider now a diatomic molecule made from two atoms which are represented as non-interacting in Fig. 4. Let Z_A and Z_B be the nuclear charge of the atoms with "core" A and B , respectively, then

$$V^A(\mu) = -Z_A/r_{A\mu} \quad (4.73)$$

$$V^B(\mu) = -Z_B/r_{B\mu} \quad (4.74)$$

and

$$V^C(\mu) = 0. \quad (4.75)$$

If we include the nuclear repulsion term, $Z_A Z_B / R_{AB}$, in the total electronic Hamiltonian, then (4.9) becomes

$$\begin{aligned} H^{A,B} = & \langle \Phi_A | \mathcal{H}_0^A | \Phi_A \rangle - \langle \Phi_A | \sum_{\mu=1}^{N_A} Z_B / r_{B\mu} | \Phi_A \rangle \\ & + \langle \Phi_B | \mathcal{H}_0^B | \Phi_B \rangle - \langle \Phi_B | \sum_{\mu=1}^{N_B} Z_A / r_{A\mu} | \Phi_B \rangle \\ & + J^{AB} - K^{AB} + Z_A Z_B / R_{AB}, \end{aligned} \quad (4.76)$$

where in (4.9) and (4.10) we took, for example,

$$\mathcal{H}^A = \mathcal{H}_0^A - \sum_{\mu=1}^{N_A} Z_B / r_{B\mu}. \quad (4.77)$$

Here $\mathcal{H}_0^A$ is the Hamiltonian of "core" A . $\mathcal{H}^B$ is similarly defined provided A and B are interchanged in (4.77).

Let us assume that the two core regions are sufficiently separated so that

$$(a) \quad E^A = \langle \Phi_A | \mathcal{H}_0^A | \Phi_A \rangle \quad (4.78)$$

and

$$E^B = \langle \Phi_B | \mathcal{H}_0^B | \Phi_B \rangle \quad (4.79)$$

represent well the "core" energies,

$$(b) \quad K^{AB} \simeq 0 \quad (4.80)$$

and

$$\begin{aligned}
 \text{(c)} \quad & Z_A Z_B / R_{AB} - \langle \Phi_A | \sum_{\mu=1}^{N_A} 1/r_{B\mu} | \Phi_A \rangle Z_B \\
 & - Z_A \langle \Phi_B | \sum_{\mu=1}^{N_B} 1/r_{A\mu} | \Phi_B \rangle + J^{AB} \\
 & \simeq (Z_A - N_A)(Z_B - N_B) / R_{AB} = z_A z_B / R_{AB}
 \end{aligned} \tag{4.81}$$

where,² for example,

$$z_A = Z_A - N_A. \tag{4.82}$$

Then we have for (4.76),

$$H^{A,B} = E^A + E^B + z_A z_B / R_{AB}. \tag{4.83}$$

Since the above assumption is compatible with the approximation used to get (4.71) and (4.72) when we considered the third choice for the "core" orbitals, then we can write the total energy in the form

$$E = \langle \Phi_C | \mathcal{H}_{\text{ALT}}^C | \Phi_C \rangle + E^A + E^B + z_A z_B / R_{AB}, \tag{4.84}$$

which is a functional of Φ_C alone. This form, as mentioned in Section IV.E, also follows if the second choice for the "core" orbitals is taken. By noting the approximations taken to obtain (4.84) we can see the validity and limitations of the use of such an expression. Szász and McGinn (1966) proposed such an equation for treating diatomic molecules without any scrutiny of its validity. They also present this with a connection to pseudo-potentials, as mentioned in Section IV.E, and, again, our purpose is their avoidance.

One might argue that the nonlocal property of our projection operators would cause difficulty. However, if one has recourse to the Roothaan expansion method such difficulties are remedied.

V. Concluding Remarks

As we stated at the beginning, our intention has been to present a review of some selected facets of the use of the projection operator technique in Hartree-Fock-type theories. However, the present article cannot be concluded without mentioning the profound and persistent influence of

² Part (c) can be broken down into its individual approximations to obtain the final form given.

a number of well-known papers by Löwdin, their scope being well beyond the framework of Hartree-Fock-theory. We would also like to point out that a recent trilogy by Adams (1967, 1969a,b) should be an excellent place to learn how profitably one can employ the technique in sophisticated Hartree-Fock-type theories.

Appendix

We shall prove here the validity of the equivalence stated in Eq. (4.38). That is, given Φ_C of Eq. (4.37), then

$$\langle \Phi_C | \mathcal{H}^C | \Phi_C \rangle = \langle \Phi_C | \mathcal{H}_{\text{equiv.}}^C | \Phi_C \rangle. \quad (\text{A.1})$$

In the following we shall suppress the index C and also we shall take for convenience, $1, 2, \dots, n = N_C$ to be the electron labels for the C region rather than $N_A + N_B + 1, \dots, N_A + N_B + N_C$.

The function Φ of Eq. (4.37) is of the form

$$\Phi = \sum_I C_I \Phi_I, \quad (\text{A.2})$$

where we have taken the summing index to capital letters for notational convenience. Here each

$$\Phi_I = A[\phi_{I(1)} \phi_{I(2)} \cdots \phi_{I(n)}] \quad (\text{A.3})$$

is a Slater determinant built from a common set of orthonormal spin orbitals and thus

$$\langle \phi_{I(k)} | \phi_{J(l)} \rangle = \delta_{I(k)J(l)}. \quad (\text{A.4})$$

Using Eq. (A.2) we have

$$\langle \Phi | \mathcal{H} | \Phi \rangle = \sum_{I,J} C_I^* C_J \mathcal{H}_{IJ}, \quad (\text{A.5})$$

where

$$\mathcal{H}_{IJ} = \langle \Phi_I | \mathcal{H} | \Phi_J \rangle, \quad (\text{A.6})$$

and, by Eq. (4.32),

$$\mathcal{H} = (1 - Q)\mathcal{H} - \mathcal{H}Q. \quad (\text{A.7})$$

Let us define, for $1 \leq k \leq n$, the orbital

$$\hat{\phi}_{I(k)}(\mu) = [1 - P_\mu^A - P_\mu^B] \phi_{I(k)}(\mu). \quad (\text{A.8})$$

Then it follows from Eq. (A.4) and the strong orthogonality conditions that

$$\langle \hat{\phi}_{I(k)} | \phi_{J(l)} \rangle = \delta_{I(k)J(l)}. \quad (\text{A.9})$$

Hence, since the antisymmetrizer, A , commutes with Q , we have that

$$(1 - Q)\Phi_I = \hat{\Phi}_I, \quad (\text{A.10})$$

where

$$\hat{\Phi}_I = A[\hat{\phi}_{I(1)}\hat{\phi}_{I(2)} \cdots \hat{\phi}_{I(n)}]. \quad (\text{A.11})$$

Upon using Eq. (A.11) and Eq. (A.7), Eq. (A.6) becomes

$$\mathcal{H}_{IJ} = \langle \hat{\Phi}_I | \mathcal{H} | \Phi_J \rangle - \langle \Phi_I | \mathcal{H} | \Phi_J \rangle + \langle \Phi_I | \mathcal{H} | \hat{\Phi}_J \rangle. \quad (\text{A.12})$$

From Eq. (4.14)

$$\mathcal{H} = \sum_{\mu=1}^n \hat{h}(\mu) + \sum_{\mu < \nu}^n 1/r_{\mu\nu}. \quad (\text{A.13})$$

By Eq. (A.9) we then have

$$\begin{aligned} \mathcal{H}_{IJ} = & \sum_k [\langle \hat{\phi}_{I(k)} | \hat{h}(1) | \phi_{J(k)} \rangle - \langle \phi_{I(k)} | \hat{h}(1) | \phi_{J(k)} \rangle + \langle \phi_{I(k)} | \hat{h}(1) | \hat{\phi}_{J(k)} \rangle] \\ & + \sum_{k < l} [\langle \hat{\phi}_{I(k)} \hat{\phi}_{I(l)} | (1/r_{12})(1 - P_{12}) | \phi_{J(k)} \phi_{J(l)} \rangle \\ & - \langle \phi_{I(k)} \phi_{I(l)} | (1/r_{12})(1 - P_{12}) | \phi_{J(k)} \phi_{J(l)} \rangle \\ & + \langle \phi_{I(k)} \phi_{I(l)} | (1/r_{12})(1 - P_{12}) | \hat{\phi}_{J(k)} \hat{\phi}_{J(l)} \rangle], \end{aligned} \quad (\text{A.14})$$

where P_{12} permutes space-spin coordinates 1 and 2. Equation (A.14) can be written as

$$\begin{aligned} \mathcal{H}_{IJ} = & \sum_k \langle \phi_{I(k)} | (1 - P_1^A - P_1^B) \hat{h}(1) - \hat{h}(1)(P_1^A + P_1^B) | \phi_{J(k)} \rangle \\ & + \sum_{k < l} \langle \phi_{I(k)} \phi_{I(l)} | [1 - (P_1^A + P_1^B)][1 - (P_2^A + P_2^B)](1/r_{12}) \\ & + (1/r_{12})[(P_1^A + P_1^B)(P_2^A + P_2^B) - (P_1^A + P_1^B) \\ & - (P_2^A + P_2^B)] | (1 - P_{12}) \phi_{J(k)} \phi_{J(l)} \rangle. \end{aligned} \quad (\text{A.15})$$

This implies that

$$\mathcal{H}_{IJ} = \langle \Phi_I | \mathcal{H}_{\text{equiv.}} | \Phi_J \rangle,$$

where $\mathcal{H}_{\text{equiv.}}$ is given in Eq. (4.38) and thus the equivalence is proved.

ACKNOWLEDGMENT

We acknowledge our debt to Professor R. Carbo for his enlightening comments on which the third section of the present article is based.

REFERENCES

- ADAMS, W. H. (1967). *Phys. Rev.* **159**, 109.
ADAMS, W. H. (1969a). *Phys. Rev.* **183**, 31.
ADAMS, W. H. (1969b). *Phys. Rev.* **183**, 37.
AUSTIN, B. J., HEINE, V., and SHAM, L. J. (1962). *Phys. Rev.* **127**, 276.
CHASE, R. L., KELLY, H. P., and KOHLER, H. S. (1971). *Phys. Rev. A* **3**, 1550.
FOCK, V., WESSELOV, M., and PETRASCHEN, M. (1940). *Zh. Eksp. Teor. Fiz.* **10**, 723.
HOUSEHOLDER, A. S. (1964). "The Theory of Matrix in Numerical Analysis." Ginn (Blaisdell), Boston, Massachusetts.
HUNT, W. J., and GODDARD, W. A., III. (1969). *Chem. Phys. Lett.* **3**, 414.
HUZINAGA, S. (1969a). *Progr. Theor. Phys.* **41**, 307.
HUZINAGA, S. (1969b). *J. Chem. Phys.* **51**, 3971.
HUZINAGA, S., and ARNAU, C. (1970). *Phys. Rev.* **1**, 1285.
HUZINAGA, S., and ARNAU, C. (1971). *J. Chem. Phys.* **54**, 1948.
HUZINAGA, S., and CANTU, A. A. (1971). *J. Chem. Phys.*, **55**, 5543.
KELLY, H. P. (1963). *Phys. Rev.* **131**, 684.
KELLY, H. P. (1964). *Phys. Rev.* **136**, B896.
LÖWDIN, P.-O. (1965). *Phys. Rev.* **139**, A357.
LYKOS, P. G., and PARR, R. G. (1956). *J. Chem. Phys.* **24**, 1166.
MCWEENY, R. (1959). *Proc. Roy. Soc. Ser. A* **253**, 242.
MULLIKEN, R. S. (1949). *J. Chim. Phys. Physicochim. Biol.* **46**, 497.
PHILLIPS, J. C., and KLEINMAN, L. (1959). *Phys. Rev.* **116**, 287.
ROOTHAAN, C. C. J. (1951). *Rev. Mod. Phys.* **23**, 69.
ROOTHAAN, C. C. J. (1960). *Rev. Mod. Phys.* **32**, 179.
SILVERSTONE, H. J., and YIN, M. L. (1968). *J. Chem. Phys.* **49**, 2026.
SZASZ, L. (1959). *Z. Naturforsch. A* **14**, 1014.
SZASZ, L., and MCGINN, G. (1966). *J. Chem. Phys.* **45**, 2898.
WEEKS, J. D., and RICE, S. A. (1968). *J. Chem. Phys.* **49**, 2741.
WILKINSON, H. J. (1965). "The Algebraic Eigenvalue Problem," Oxford Univ. Press, (Clarendon), London and New York.